The Sun and the Inner Planets

The Sun is a G-type main-sequence star that contains about 99.8 percent of the total mass of the solar system. Light from the Sun takes about 8 minutes and 20 seconds to reach Earth. Mercury is the smallest planet and the closest to the Sun; a year on Mercury lasts only 88 Earth days. Venus is the hottest planet, with surface temperatures around 465 degrees Celsius, because its thick carbon dioxide atmosphere traps heat in a runaway greenhouse effect. Earth is the only planet known to support life, and about 71 percent of its surface is covered by water. Mars is called the Red Planet because iron oxide (rust) covers its surface. Mars is home to Olympus Mons, the tallest volcano in the solar system, about 22 kilometres high.

The Gas Giants and Ice Giants

Jupiter is the largest planet; more than 1,300 Earths could fit inside it. Its Great Red Spot is a storm larger than Earth that has lasted for at least 350 years. Jupiter has 95 officially recognised moons, including Ganymede, the largest moon in the solar system. Saturn is famous for its bright rings, which are made mostly of ice particles with some rock and dust. Saturn's density is so low that it would float in water. Uranus rotates on its side, with an axial tilt of about 98 degrees, probably caused by a giant collision long ago. Neptune is the windiest planet, with winds reaching more than 2,000 kilometres per hour. Neptune was the first planet found by mathematical prediction rather than by direct observation, in 1846.

Beyond the Planets

The asteroid belt lies between Mars and Jupiter. Its largest object, Ceres, is classified as a dwarf planet. Beyond Neptune lies the Kuiper Belt, a region of icy bodies that includes Pluto. Pluto was reclassified from a planet to a dwarf planet by the International Astronomical Union in 2006. Comets are icy bodies that grow glowing tails when they approach the Sun; Halley's Comet returns roughly every 76 years and will next be visible from Earth in 2061. The Voyager 1 spacecraft, launched in 1977, entered interstellar space in 2012 and is the most distant human-made object.